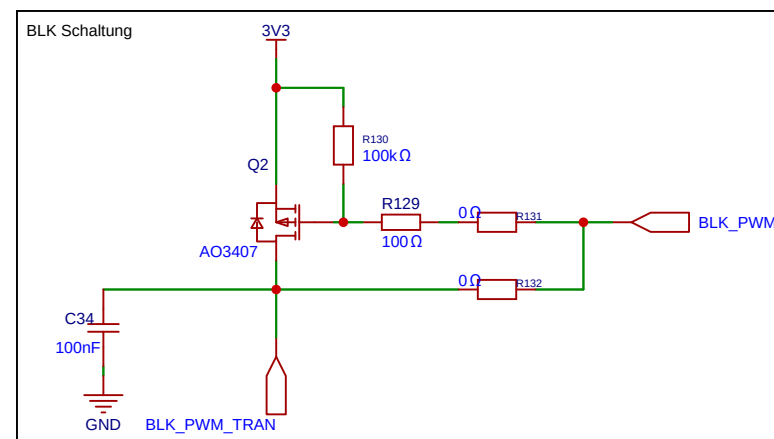
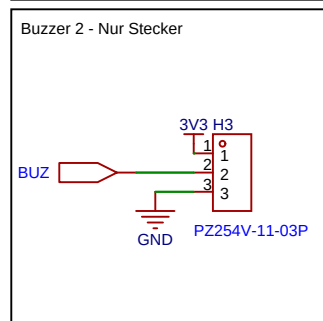
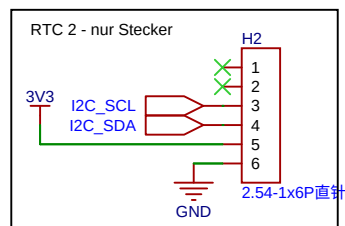
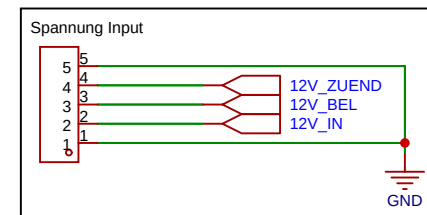
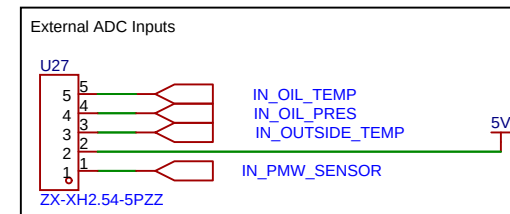
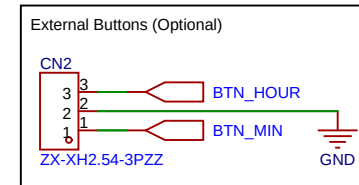
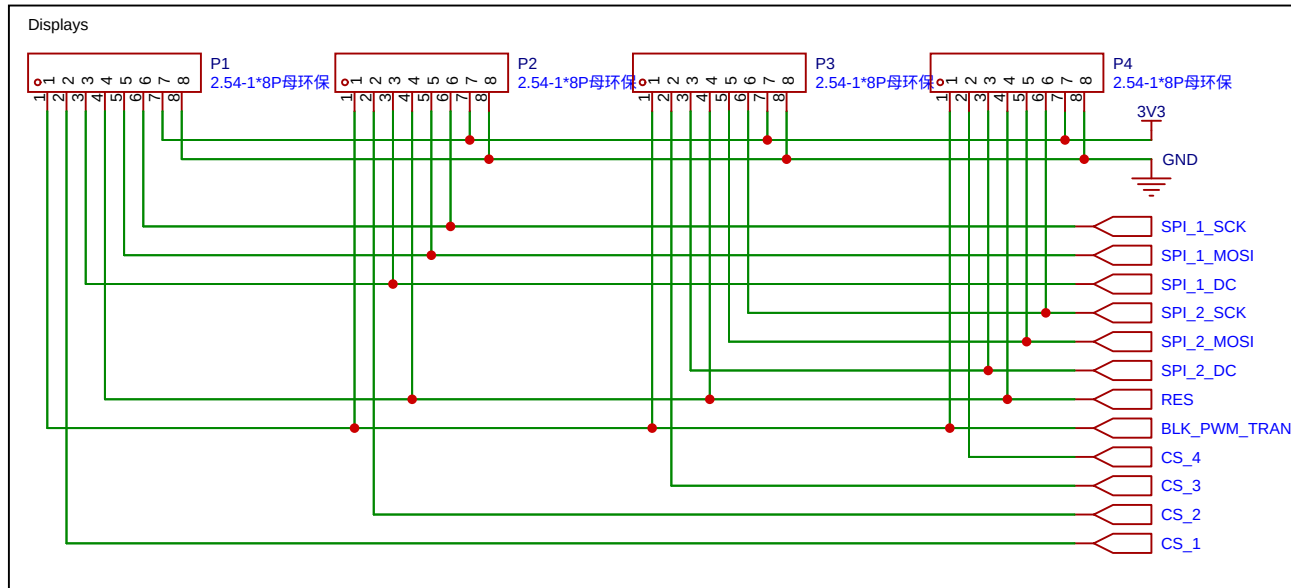
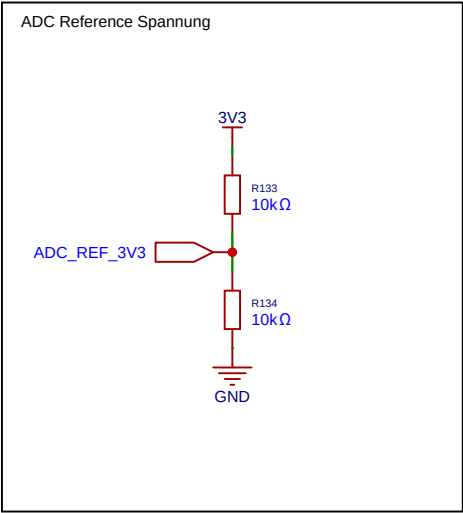
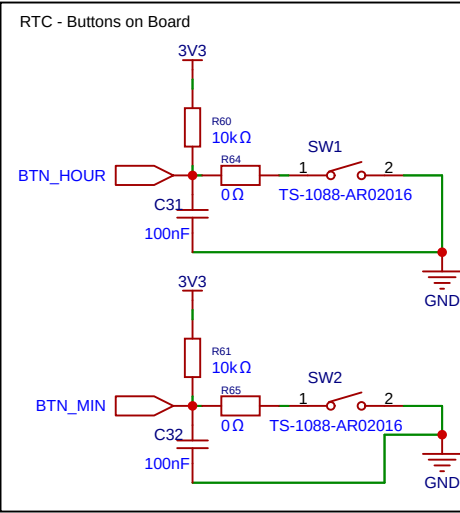
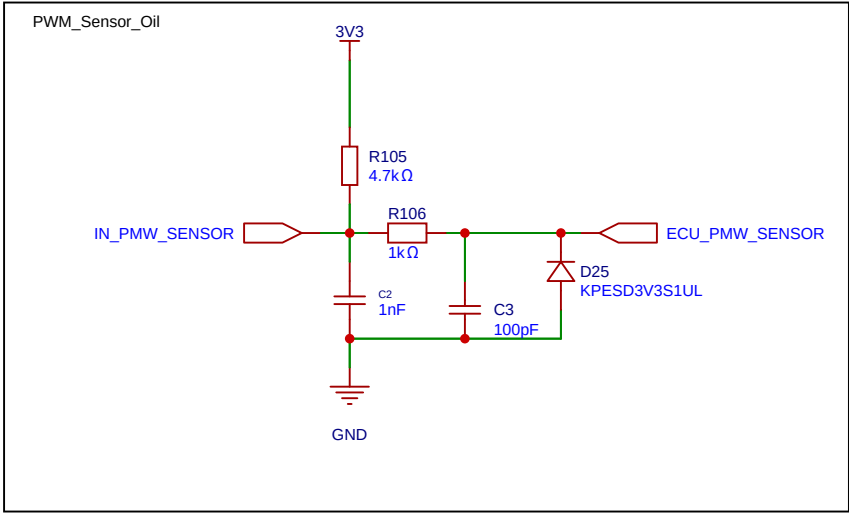




Schematic	V4			Create at	2026-05-06
Board	V4			Update at	2026-05-06
Drawn				Page	Displays und Connections
Reviewed				PKW_Display_V4_Breit_V1	
		Version	Size	Page 2 Total 5	
		V1.0	A4	EasyEDA.com	



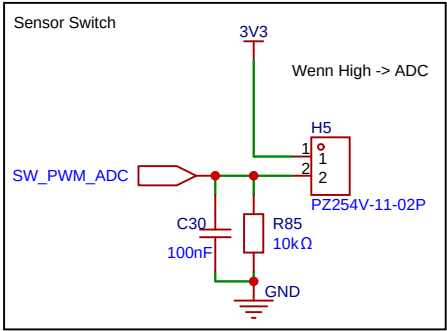
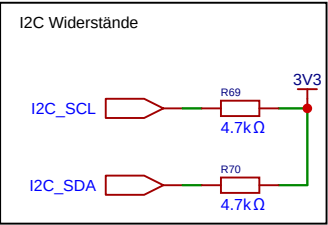
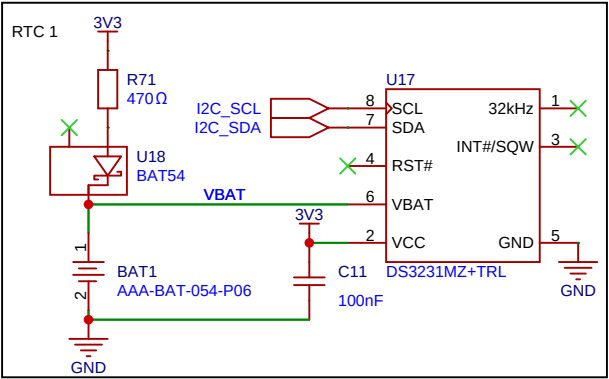
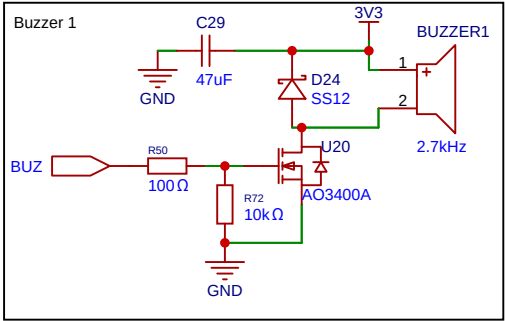
ToDo Rev 2: Spannungsmessung von 3.3V hinzufügen und in Code einbauen

3.3V
|
10k
|
ADC
|
10k
|
GND

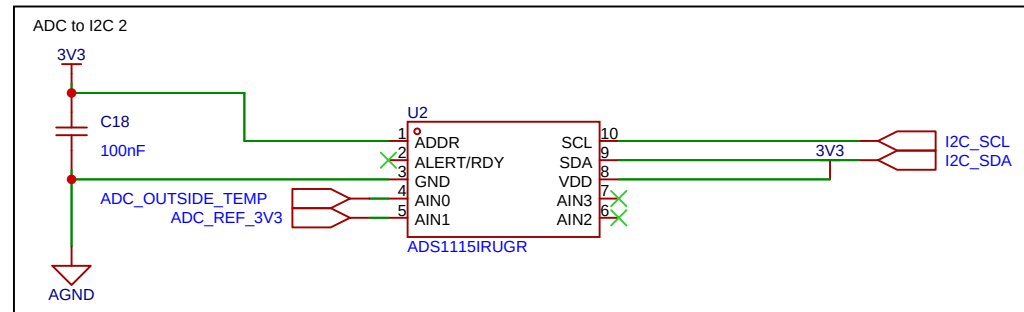
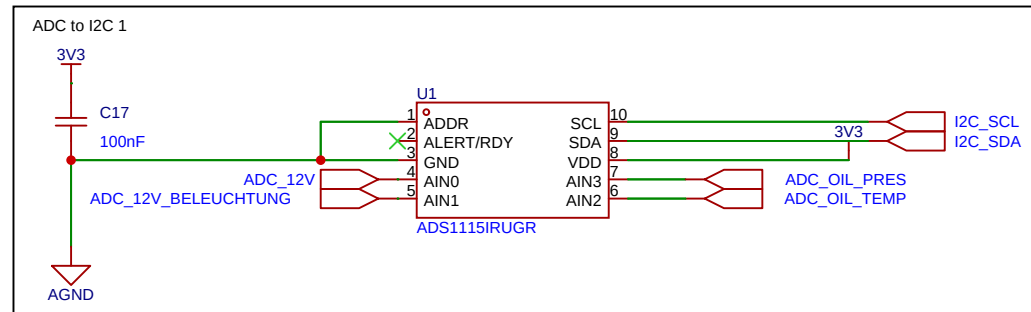
Nutzen für 12V Messung

Den Pullup möglichst nahe am ESP platzieren.
Nicht am Sensorstecker.
1 C3 so nah wie möglich an den ESP-Pin
2 Pullup ebenfalls nahe am ESP
3 Die Leitung vom Stecker bis R106 möglichst nicht parallel zu:

Buck-Induktivität
SW-Pin vom LMZM
SPI-Leitungen



Schematic	V4	Create at	2026-05-06
		Update at	2026-05-06
Board	V4	Page	RTC, Buzzer, PWM, I2C
Drawn	PKW_Display_V4_Breit_V1		
Reviewed			
		Version	Size
		V1.0	A4
		Page 3 Total 5	
		EasyEDA.com	



Wenn du zwei ADS1115IRUGR verwendest:

Lege alle ADC-Filterkondensatoren wirklich nahe an die ADC-Pins.

AGND-Fläche kompakt halten.

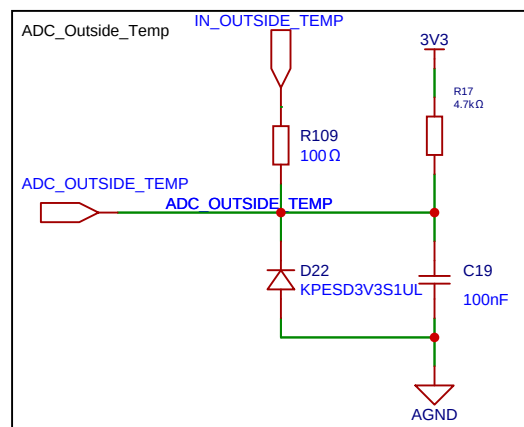
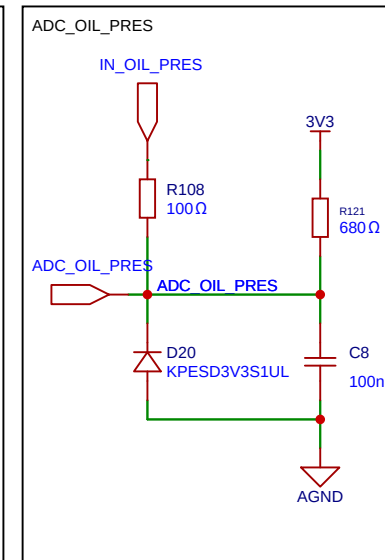
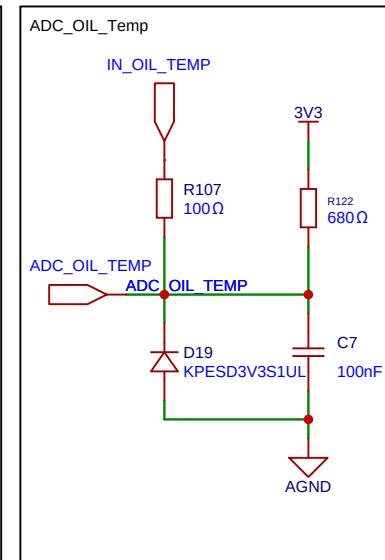
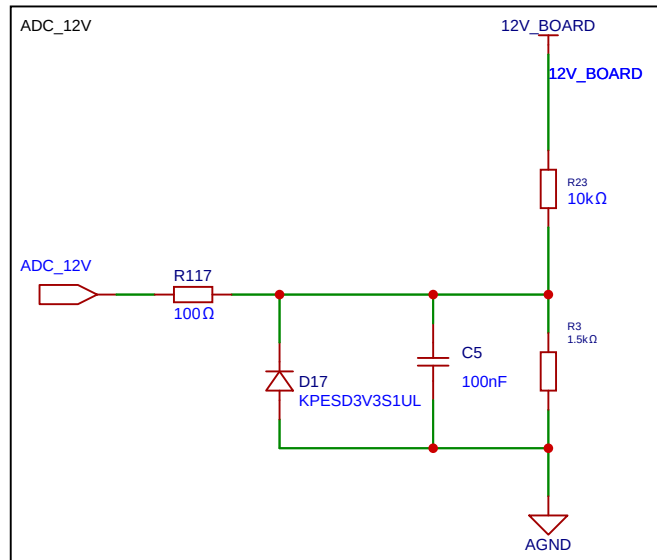
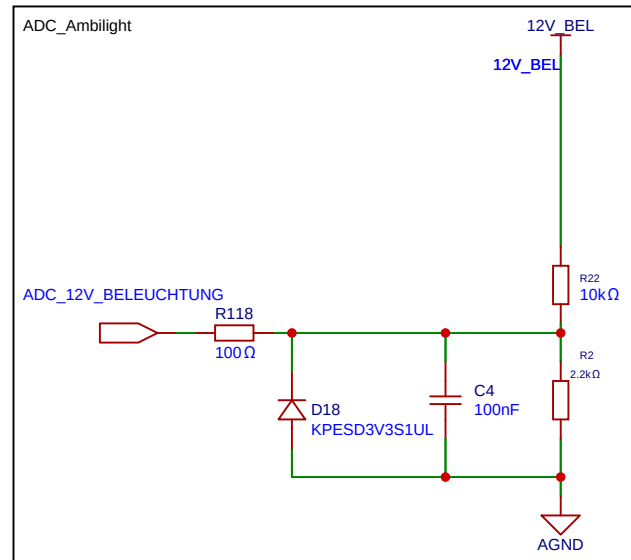
Serienwiderstände am Stecker platzieren.

TVS möglichst nahe am Eintrittspunkt.

So bekommst du saubere Messwerte auch bei laufendem Motor.

Stecker → Teiler / Pullup → RC → 100Ω → ESP-Pin

ESD-Diode



Schematic	V4			Create at	2026-05-06
				Update at	2026-05-06
Board	V4			Page	ADCs
Drawn		PKW_Display_V4_Breit_V1			
Reviewed					
		Version	Size	Page 4 Total 5	
		V1.0	A4	EasyEDA.com	

